

Blacksburg, VA
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Pratheek Prakash Shetty

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"Systems engineer with experience in compiler optimization, system software, and hardware acceleration, specializing in performance optimization and parallel computing."

SKILLS

- Proficient in: Python, C++, C, Java, JavaScript, Rust, Go, TypeScript, SQL, RabbitMq, Scala
 - Parallel Computing: CUDA, OpenMP, Multithreading, High-Performance Computing (HPC)
 - Fundamentals: Data Structures & Algorithms, UNIX, Linux, Operating Systems, Computer Architecture
 - ML/Compilers: Pytorch, TensorFlow, LLVM IR, Data Flow Analysis, MLIR, Compiler Optimisation
 - Web Development: Docker, React, MongoDB, PostgreSQL, AWS (EC2, S3, Lambda), Django
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EDUCATION

Master of Science, Computer Engineering Aug 2024 - May 2026 (Expected)
Virginia Polytechnic Institute and State University

- Coursework: Compiler Optimization, Computer Vision, Computer Architecture, Multi-processor programming, Advanced Parallel Computation.

Bachelor of Engineering, Computer Science and Engineering Aug 2019 - Aug 2023
Acharya Institute of Technology, Bengaluru

RESEARCH EXPERIENCE

Research Assistant, Synergy Lab, Virginia Tech Dec 2024 - Present

- **GPU Atomics Characterization** - Analyzed atomic operation performance across modern NVIDIA, AMD and Intel GPUs using architecture specific profiling tools (Nsight Compute, rocprof, VTune); studied contention patterns, memory coherence, and concurrent data structure behavior across different GPU architectures.
 - **Wait-Free Concurrent Queue Design** - Implemented and optimized wait-free queue in C++; profiled and eliminated 75% malloc bottleneck through static pre-allocation, achieving $31\times$ memory overhead reduction; developed unified memory hybrid design and cache-aware variant for AMD MI300A unified memory.
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PROFESSIONAL EXPERIENCE

Software Engineer, Virginia Tech Sep 2024 - May 2025

- Building a real-time data acquisition system to enable researchers to capture high-frequency signals (10MHz+) using Electron, React and Python.
- Implemented multi-threaded data processing pipeline, increasing throughput by 2.5x.
- Reduced processing latency by 35% by optimizing microcontroller communication with custom SPI.

SDE Intern, Instahyre Mar 2024 - Jun 2024

- Restructured microservices architecture for 3x more concurrent users, reducing API response time by 40%.
 - Optimized the PostgreSQL queries and Django Backend, decreasing average query time from 1.2s to 0.3s.
 - Developed automated personalized email template system that increased candidate response rates by 22%.
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PROJECTS

Autotiling Pass for LLVM/MLIR March 2025 - May 2025

- Built a C++ MLIR pass that analyzes data reuse patterns, selects cache-optimal tile sizes, fuses loops, and inserts prefetches for GPU kernels; achieved $1.9\times$ speedup on GEMM/Conv with 2% compile overhead

RISCV Sine Hardware Accelerator Nov 2024 - Dec 2024

- Designed custom RISCV hardware accelerator for trigonometric functions to address computational bottlenecks in graphics engines, physics engines etc.
- Achieved 43% reduction in cycle count compared to software implementation while maintaining precision.

Rlu-BplusTree: BPlusTree in rust for performance and safety Sep 2024 - Dec 2024

- Created concurrent B+ tree implementation to address data race issues in multi-threaded database systems.
- Built to be used for Databases and file systems with better performance against other synchronization methods.
- Achieved 90% of C implementation speed while eliminating memory safety vulnerabilities present in C version.